FROM COPILOT TO CO-THINKER: SITUATING LLMS IN THE DESIGNER'S EVOLVING THINKING FLOW

Lu ZHANG[1], **Xiaobo QIAN**[1]
[1]Jiangnan University, China

ABSTRACT
The integration of Large Language Models (LLMs) into design practices is often narrowly focused on efficiency, treating LLMs as a mere "pilot" for executing tasks. This study proposes a perspective shift, advocating for a more nuanced, situational approach where LLMs are positioned as "co-thinkers" within the designer's evolving thinking flow. Through a comparative analysis of three design cases, encompassing brand strategy (Fuguang), product innovation design (Robam), and user experience research (Little Swan), we elaborate on how the roles of both designers and LLMs dynamically shift across different design contexts and stages of the design thinking process. Our findings reveal a spectrum of collaborative modes: LLMs act as a macro-scanner in strategic research, a creative provocateur in product conceptual design, and a reflective partner in experience improvement. Correspondingly, designers evolve from tool operators to guides of reflection. Furthermore, a dynamic designer-LLMs collaboration framework is proposed, leveraging LLMs' generative potential while navigating ethical considerations. This framework moves beyond a one-size-fits-all model, inspiring designers to maintain their critical leadership and act as multipliers in the age of LLMs.

Keywords: Designer-LLMs Collaboration, Design Thinking Model, Case Study

1 INTRODUCTION

Large Language Models (LLMs) have emerged as a pivotal social technology, capable of performing tasks such as natural language understanding, intention recognition, reasoning, and language generation, enabled by vast parameter counts and deep learning architectures [1]. Their ability to engage in open dialogue, akin to human thinking, has been applied to various design domains, such as industrial production [2], robot design [3], and user experience design [4]. By supporting designers decision-making [5] and fostering innovative solutions [6], LLMs are increasingly integrated into the design process.

While both academia and industry are actively exploring how designers can collaborate with AI , current practices predominantly focus on integrating AI tools into design workflows to improve efficiency and productivity. The dominant approach remains the copilot model, which emphasizes automation and operational support. For instance, research by the Microsoft team has demonstrated that LLMs can significantly enhance workflow efficiency [7]. In this mode, LLMs assist in collecting user data, analyzing behavioral patterns, and generating user insights [8]. LLMs can identify user preferences and pain points from feedback through techniques such as assisted-rooted coding [9] and text cluster analysis [10]. Although effective in accelerating specific design tasks, this model largely treats LLMs as functional tools, a perspective that overlooks their potential for deeper, thinking-oriented collaboration. To move beyond this instrumental view, this paper argues for a shift from copilot to co-thinker, positioning LLMs as active participants within designers' thinking processes. With strong generative and analytical capabilities, LLMs can effectively inspire designers and foster reflection during early-stage conceptual exploration. Through a comparative analysis of three design cases of brand strategy, product innovation, and user experience research, we investigate how designers and LLMs collaboratively interact across different stages of the design thinking model. By synthesizing these practical insights into a structured framework, this research aims to maintain designers' critical leadership while leveraging LLMs' generative potential.

2 LITERATURE REVIEW

2.1 LLMs in Design：From Tools to Partners

Since the emergence of the GPT series in 2023, design researchers have increasingly recognized the significant advantages of Large Language Models (LLMs) in areas such as text generation, information extraction, semantic understanding, and data analysis, leading to their adoption as auxiliary tools in design research. On one hand, LLMs are employed to collect user data, enabling more accurate user analysis and deeper insights. For instance, by analyzing user comments and social media data, LLMs can identify user preferences and pain points regarding product features, aesthetics, and interactions. On the other hand, LLMs facilitate the semi-automated analysis of this data. Methods such as assisted grounded coding [11] and text cluster analysis [12] significantly enhance efficiency and reduce repetitive manual labor.

With the rise of human-AI collaboration concepts (e.g., More-than-human design), a growing number of researchers have recognized that the potential of LLMs extends to reasoning and self-optimization. These advanced capabilities allow LLMs to assist designers in refining conceptual prototypes [4] and accelerating design iteration cycles [9][13]. Studies have demonstrated that LLMs can support wiser user decision-making through conversational reflection [14], help designers mitigate cognitive biases during the design process [15], and create opportunities for their integration into design decision-making [5]. From a process perspective, the involvement of LLMs has expanded from late-stage execution to early-phase exploration, such as information categorization and goal setting during user research. Relevant studies also emphasize that design principles for generative AI applications should support a more active role for LLMs in preliminary exploration.

2.2 Evolving Roles: Interactions between Designer and LLMs

The role of LLMs has evolved from a mere tool to an assistant, fostering new forms of interaction between designers and AI. Existing research has begun to explore these dynamic relationships across various design contexts. For example, Subramonyam et al. conceptualize the Designer-LLMs interaction as a prompt-based, iterative cycle, positioning the user-researcher-in-the-loop [16]. Titta Jylkäs [17] proposed a thinking process for designer-AI collaborative service design, while others, like Li et al. [18] have investigated non-linear collaboration between humans and AI agents in UX design. Researchers like Li Xu et al. [15] have explored co-creation models with AIGC, describing it as a dual-track parallel process. Whether driven by theory or derived from practice, these discussions fundamentally address the core of design thinking. Design thinking is a human-centered methodology for tackling complex problems, characterized by its cyclical and iterative nature. The design focus, encompassing user needs, value objectives, and design directions, evolves from an initial state of ambiguity to one of increasing clarity.

However, current research on the relationship between LLMs and designers often remains at a macro-level discussion of AIGC-human collaboration or focuses narrowly on specific tools and singular scenarios. There is a lack of systematic comparison of collaborative patterns between LLMs and designers across different design contexts. Furthermore, within these new interactive relationships, designers must also consider the ethical implications and risks associated with LLMs. Current limitations include weak emotional and situational understanding, hallucinations, data biases, and privacy leaks [19]. For instance, LLMs can perpetuate cultural, ethnic, or linguistic stereotypes when generating user profile descriptions [20]. Therefore, effective collaboration [21] hinges on establishing a complementary partnership between designers and LLMs.

2.3 Exploring Designer-LLMs Collaboration mode

Against this backdrop, this study proposes a dynamic Designer-LLMs Collaboration Framework (Figure 1), built upon the foundational Double Diamond model. The Double Diamond model divides the design process into four key stages: Discover, Define, Develop, and Deliver. Through its two cycles of divergence and convergence, it ensures that "the right problem is addressed" and "the problem is solved right." This framework has been widely adopted in service design, user experience, industrial design, and business strategy, establishing itself as a fundamental model for innovation. Our proposed collaboration process aligns with these four stages. Designers set the objectives, utilize LLMs to gather and organize information, and then evaluate and refine the analysis and insights generated by the LLMs.

Based on the design direction derived from this process, designers further leverage LLMs to inspire multidimensional creativity, screen conceptual ideas, and ultimately produce design solutions.
Existing research largely confirms this complementary relationship but often adopts a static, task-oriented perspective, focusing on the capabilities of LLMs in discrete tasks. A critical gap remains in understanding this collaboration as a dynamic, role-shifting partnership that evolves throughout the entire design thinking process. This study aims to address this gap by exploring the following research question: How can classical design thinking models be adapted to frame LLMs not merely as tools, but as flexible "co-thinkers"?

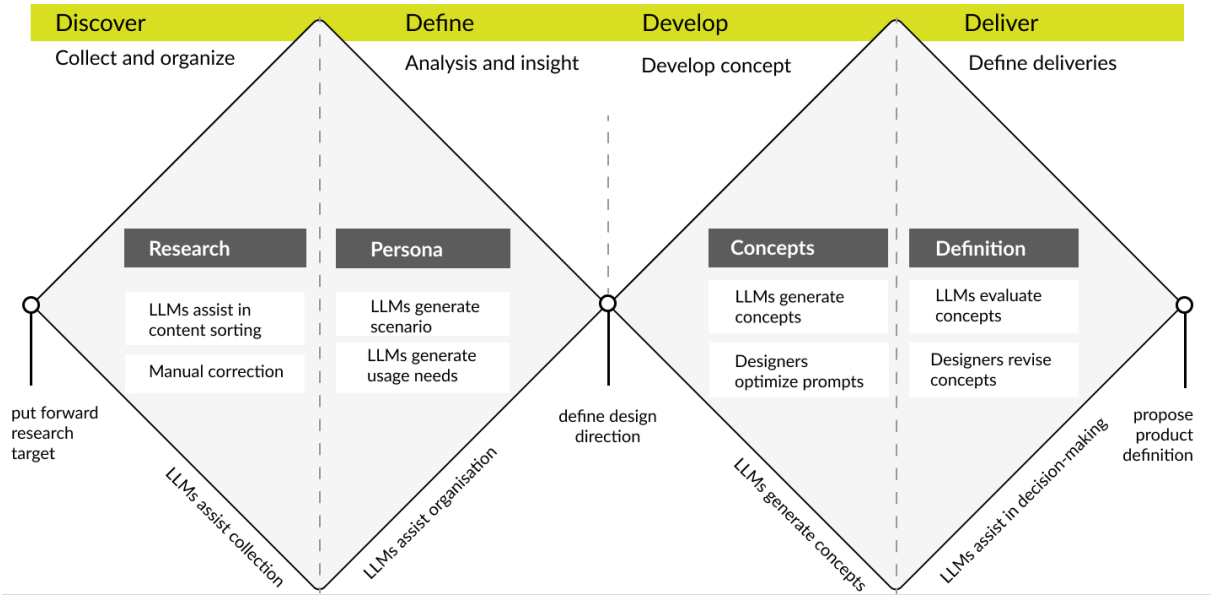


Figure 1. Hypothesis of Designer-LLMs Collaboration Framework.

This leads to the following research questions:

- RQ1: Across different types of design projects, what primary thinking roles do LLMs assume?
- RQ2: How do designers' core responsibilities dynamically adjust in response to the evolving roles of LLMs?
- RQ3: What principles can guide designers in maintaining critical leadership throughout this collaborative process?

3 METHODOLOGY: A MULTI-CASE STUDY APPROACH

This study employs a qualitative multi-case study approach to investigate the dynamic collaboration between designers and LLMs across diverse design contexts. This methodology is particularly suited to capturing the nuanced, making contextualized comparison of how Designer-LLMs collaboration varies across scenarios.
Three distinct design projects were selected to form a comparative foundation:

- Fuguang Coffee Cup (Overseas Market Research): Representing the context of brand strategy, characterized by macro-level uncertainty and exploratory research.
- Robam Appliance (Kitchen Island Solution Design): Exemplifying product innovation, focusing on breakthrough concept generation.
- Little Swan (User Experience Research): Centering on user experience of washing machine, aimed at the deep optimization of an existing user journey.

Data Collection and Analysis

The primary data for analysis comprised detailed project documentation, including prompt histories, intermediate LLMs-generated outputs, and final design delivers. The analysis followed a structured, comparative process focusing on the core chain of:

- Design Challenge: Identifying the specific design challenge at each stage of the process.
- LLMs Behavior: Cataloging LLMs behaviors, including the functions it performed (e.g., data summarization, idea generation) and the nature of its outputs.
- Designer Intervention: Analyzing designer interventions, such as how prompts were iteratively

refined, how LLMs outputs were critiqued, integrated, or overruled, and what strategic decisions were made.

- Collaborative Output: Evaluating the final collaborative output that resulted from this interaction.

By comparing these chains across the three cases, we were able to identify patterns in how the roles of both the designer and the LLMs shifted in response to different design contexts and stages of the design thinking process.

4 CASE STUDY

4.1 Case 1: Fuguang Coffee Cup Market Research

Design Challange

Fuguang, a Chinese local thermal cup brand, faced the challenge of entering the global coffee cup market, which is characterized by diverse and deeply entrenched cultural practices. The primary challenge was the macro-level uncertainty of navigating vastly different coffee cultures in Italy, North America, and Southeast Asia to define a coherent brand and product strategy.

LLMs as a Macro-Scanner & Cultural Analyst

In this strategic research phase, the LLM was primarily deployed as a macro-scanner to rapidly synthesize a vast amount of information. Its key behaviors included:

- Cultural Pattern Identification: Analyzed and summarized cultural texts, social media trends, and market reports to identify core coffee consumption rituals, values, and unmet needs in each target region.
- Strategic Insight Generation: Compared and contrasted findings across markets to highlight global commonalities and critical local differences.

Designer as Synthesizer

Designers intervened by moving beyond mere prompt operation to guide the analytical process and inject strategic interpretation. Key interventions were:

- Framing the Inquiry: Crafted iterative prompts to steer the LLMs' analysis from broad cultural overviews to specific product-related behaviors.
- Critical Synthesis: Critically evaluated the LLM-generated insights, identifying potential biases or oversimplifications, and integrated them with prior business knowledge to form a holistic view.
- Strategic Decision-Making: Translated the synthesized insights into a concrete brand positioning and product strategy report, made final decisions on market prioritization and value proposition.

Collaborative Output

The collaboration yielded a comprehensive brand and product strategy report that clearly articulated how Fuguang could adapt its offerings to resonate with distinct coffee cultures while maintaining a unified global brand identity.

4.2 Case 2: Robam Kitchen Island Conceptual Design

Design Challange

Robam aimed to enhance innovation in its kitchen appliance line. The challenge was one of breakthrough concept generation—moving beyond incremental improvements to identify novel and effective ways to integrate LLMs into the early-stage conceptual design of products like a kitchen island.

LLM as a Creative Provocateur

In the conceptual development stage, the LLM's role shifted to that of a creative provocateur. Its key behaviors included:

- Divergent Ideation: Rapidly generated a wide range of product concepts, features, and user scenarios for kitchen islands, pushing beyond conventional ideas.
- Functional Integration: Proposed novel user interaction within the kitchen ecosystem.

Designer as Concept Curator

Designers leveraged the LLM's generative capacity while maintaining critical creative leadership. Their interventions included:

- Refining Prompts: Used workshop discussions with corporate designers to define creative boundaries and then crafting prompts to guide the LLM's ideation.
- Curation: Filtered and combined the LLMs - generated concepts based on technical feasibility, user desirability, and business viability.

Collaborative Output

The collaboration resulted in a portfolio of innovative, LLMs - augmented product concepts for a kitchen island solution, which were communicated and refined into actionable proposals for further development.

4.3 Case 3: Little Swan Washing Machine UX Research

Design Challange

The project focused on deep user experience optimization for Little Swan washing machines. The challenge was to synthesize insights from over 236,000 words of user interviews, covering diverse family structures like single occupants, families with infants, and multi-generational households—to identify nuanced pain points and generate innovative, feasible product concepts that truly addressed user needs.

LLMs as an Efficient Scanner and Reflective Partner

The LLMs' role dynamically evolved across the design thinking stages:

- In Discover: Functioned as a scanner, rapidly structuring vast qualitative data into organized demand analyses (e.g., categorizing needs into functionality, performance, and design) within minutes, significantly boosting efficiency.
- In Define: Acted as a tool and reflective partner, generating user persona drafts and iteratively helping to identify pain points through seven rounds of dialogue.
- In Develop: Served as a creative stimulator. In one mode, it refined designer-proposed concepts; in another, it generated initial concepts itself. Notably, its occasional "hallucinations" provided unexpected, creative perspectives.

Designer as an Analyst

Designers maintained critical leadership by intervening at strategic points:

- Guiding Analysis: Crafted prompts to steer the LLMs' analysis and manually filtered its conventional or technically generic outputs (e.g., "high-temperature drying causes wrinkles") to uncover truly innovative opportunities.
- Curating Ideas: Evaluated, selected, and iterated on LLMs-generated concepts, focusing on feasibility and differentiation from existing products.

Collaborative Output

Based on the analysis of user interview transcripts, designer generated four distinct family-based user personas. While these personas carry certain stereotypical characteristics, they efficiently distinguished the primary user segments and core usage contexts for Little Swan washing machines. This process yielded key experiential optimization insights, notably the need for "more precise and customizable control modes" and the potential for "modular, upgradable product configurations."

4.4 Cross-Case Comparison

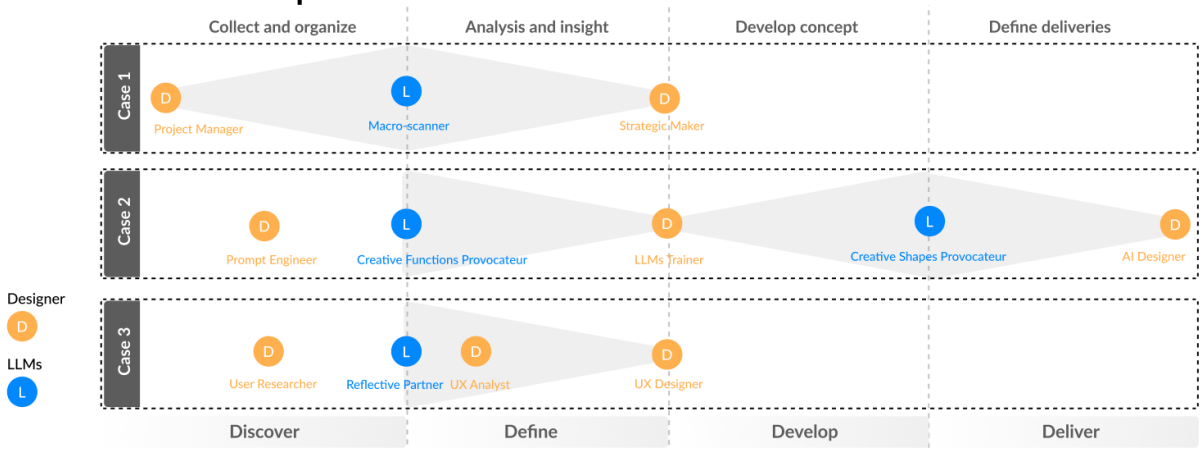


Figure 2. Collaboration Mode Comparison.

The three case studies reveal that the role of LLMs is not fixed but dynamically co-evolves with the designer's role across different design contexts and thinking stages. As Figure 2 depicts, in the strategic research for Fuguang (Case 1), LLMs serve as a Macro-scanner; in the product innovation for Robam (Case 2), as a Creative Provocateur; and in the UX refinement for Little Swan (Case 3), as a Reflective Partner. Another key finding is that LLMs excel in the early, divergent phases of design thinking (Discover, Develop), where speed, scale, and generative capacity are paramount.

Reflections on how to better utilize LLMs in projects:

1. Before employing LLMs for analysis, it is essential to build an expert knowledge base so that LLMs first grasp domain-specific norms and professional knowledge—condensed from real-world human experiences. Only then can LLMs better fulfil designers' requirements.
2. LLMs facilitate mixed-methods research, particularly exploratory approaches — e.g., using qualitative data to interpret quantitative results, or using quantitative data to generalize qualitative findings.

4 DISCUSSION

The evolving thinking process resembles a dynamic, double-loop dialogue where designers and LLMs interact synergistically, advancing through design thinking stages (Figure 3) .

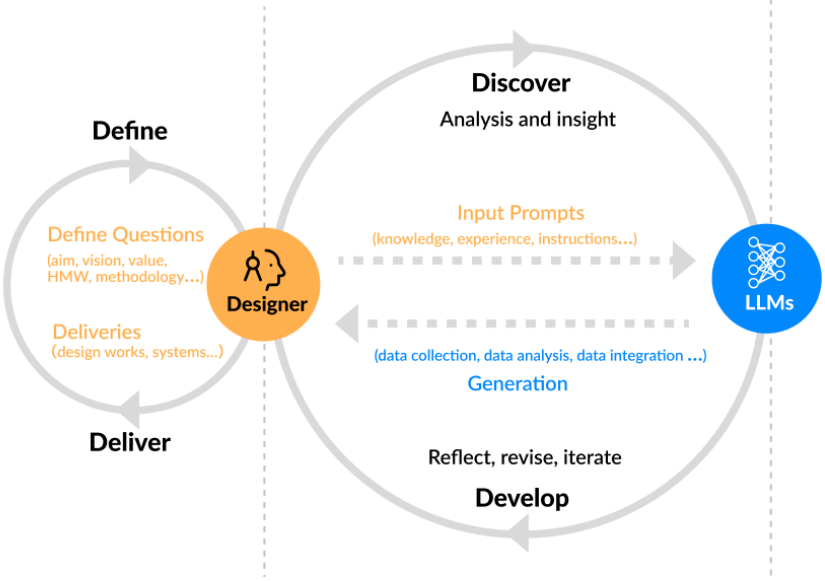


Figure 3. The Designer-LLMs Evolving Thinking Flow.

RQ1: The Evolving Thinking Roles of LLMs

During the Discover and Develop stage, LLMs, as an efficient data collection and organization tool, enhance research efficiency, enabling designers to focus on more integrated and insightful analytical

work. During the Define and Deliver stage, LLMs assist designers in conducting qualitative data analysis of users, helping them quickly extract key requirements from the chaotic user demands and identify valuable design opportunities.

RQ2: The Evolving Thinking Roles of Designers

While LLMs provide strong computational capabilities — enabling high-speed, large-scale data processing as well as the execution of both repetitive and creative generative tasks. Designers play a decisive role in design thinking process. Whether it involves planning and management of the whole project, providing LLMs with sufficient context and guidance, critically evaluating and adopting LLMs' suggestions during the mid-phase, or finalizing refinements and deliveries in the later stages, the judgment and leadership of designers are evident throughout the collaboration.

RQ3: Principles for Maintaining Designers' Leadership

The collaborative flow necessitates that designers maintain cognitive agency to navigate the inherent "black-box" nature of LLMs. Our analysis surfaced several critical considerations for effective collaboration:

1. Treating LLM Output as Creative Stimulus, Not Final Solution: LLMs are unable to create new knowledge and are better at incremental innovation, so the generated innovative concepts are more often used as materials to stimulate creativity rather than as achievements that can be directly borrowed.
2. The Critical Role of Contextual Richness: The output rationality of LLMs is relatively weak, designers should provide sufficient contextual information prompts and avoid excessive reliance.
3. Proactive Mitigation of Inherent Biases: As LLMs' training datasets may contain biases and stereotypes, designers must avoid using leading or suggestive language in communication.
4. Prioritizing Precision to Counteract Hallucinations: To mitigate hallucinations, designers should ensure that the prompt words are as precise as possible rather than vague, avoid junk input and junk output.
5. Verifying Nuanced and Emotional Insights: LLMs have difficulty identifying the emotions and situations and may miss some subtle yet crucial insights, which requires designers to verify and refine.

6 CONCLUSION

This study examines the evolving thinking flow between designers and LLMs across different design stages. Findings confirm that the collaboration is a fluid, evolving partnership rather than a static, tool-based interaction. In the age of LLMs, the designer's role transforms from a hands-on executor to a strategic multiplier. The LLMs act as a force amplifier, but the designer's critical thinking, ethical judgment, and creative vision serve as the essential multiplier at the source. We hope this preliminary exploration inspires further research into collaborative models between designers and LLMs.

REFERENCES

[1] Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., *et al.* A survey of large language models. *arXiv preprint*, 2023. Available online: https://arxiv.org. Accessed 13 April 2025.

[2] Makatura, L., Foshey, M., Wang, B., Hähnlein, F., Ma, P., Deng, B., *et al.* An MIT exploration of generative AI: large language models for design and manufacturing. 2023. Available online: https://mit-genai.pubpub.org.

[3] Stella, F., Della Santina, C. and Hughes, J. How can LLMs transform the robotic design process? *Nature Machine Intelligence*, 2023, 5(6), 561–564.

[4] Ekvall, H. and Winnberg, P. *Integrating ChatGPT into the UX Design Process: Ideation and Prototyping with LLMs*. Master's dissertation, Umeå University, Department of Informatics, 2023.

[5] Kim, D. H., Jeong, D., Yadgarova, S., Shin, H., Son, J., Subramonyam, H., and Kim, J. PlanTogether: facilitating AI application planning using information graphs and large language models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2025, pp. 1–23.

[6] Figoli, F. A., Mattioli, F. and Rampino, L. *Artificial Intelligence in the Design Process: The Impact on Creativity and Team Collaboration*. OAPEN, 2022. Available online: https://library.oapen.org.

[7] Cambon, A., Hecht, B., Edelman, B., Ngwe, D., Jaffe, S., Heger, A., *et al.* Early LLM-based tools for enterprise information workers likely provide meaningful boosts to productivity. Microsoft Research, 2023. Available online: https://www.microsoft.com. Accessed 20 December 2024.
[8] Yang, H. The integration of data and design: the innovation path research of user requirement insight through big data analysis. *Zhuangshi (Decoration)*, 2019, (5), 100–103.
[9] Sinha, R., Solola, I., Nguyen, H., Swanson, H. and Lawrence, L. The role of generative AI in qualitative research: GPT-4's contributions to a grounded theory analysis. In *Proceedings of the Symposium on Learning, Design and Technology*, 2024.
[10] Ganwani, S., Ratkal, S., Mahamuni, R. and Pathak, I. Synergizing human expertise with AI: the role of LLMs in user research. In *Proceedings of the Design Research Society Conference (DRS 2024)*, Boston, USA, June 2024.
[11] Guo, J., Yin, Y., Sun, L. and Chen, L. Empirical study of problem–solution co-evolution in human–GAI collaborative conceptual design. In *Proceedings of the Design Research Society Conference (DRS 2024)*, Boston, USA, June 2024.
[12] Muhs, N. and Stankowski, A. Leveraging LLMs for reflection: approaches to mitigate assumptions within the design process. In *Proceedings of the Design Research Society Conference (DRS 2024)*, Boston, USA, June 2024.
[13] Musi, E., Kokciyan, N., Al-Khatib, K., Ceolin, D., Dietz, E., Gutekunst, K., *et al.* Toward reasonable parrots: why large language models should argue with us by design. *arXiv preprint*, 2025.
[14] Subramonyam, H., Thakkar, D., Ku, A., Dieber, J. and Sinha, A. K. Prototyping with prompts: emerging approaches and challenges in generative AI design for collaborative software teams. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2025, pp. 1–22.
[15] Li, Z., Zhu, H., Lu, Z., Xiao, Z. and Yin, M. From text to trust: empowering AI-assisted decision making with adaptive LLM-powered analysis. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2025.
[16] Jylkäs, T. *Service Design and Artificial Intelligence in Designing Human-Centred Digital Services*. Doctoral dissertation, Acta Electronica Universitatis Lapponiensis, 2020.
[17] Zhou, J., *et al.* Understanding nonlinear collaboration between human and AI agents: a co-design framework for creative design. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2024, pp. 1–16.
[18] Li, X., Liu, Y. and Huang, G. Research on man–machine co-creation design of intangible cultural heritage digital creative products from the perspective of AIGC multimodal narrative. *Packaging Engineering*, 2025, 1–14.
[19] Watkins, R. Guidance for researchers and peer-reviewers on the ethical use of large language models in scientific research workflows. *AI and Ethics*, 2023, 4(4), 969–974.
[20] Sattele, V. and Ortiz, J. C. Generating user personas with AI: reflecting on its implications for design. In *Proceedings of the Design Research Society Conference (DRS 2024)*, Boston, USA, June 2024.
[21] Song, B., Zhu, Q. and Luo, J. Human–AI collaboration by design. In *Proceedings of the Design Society*, 2024, 4, 2247–2256.